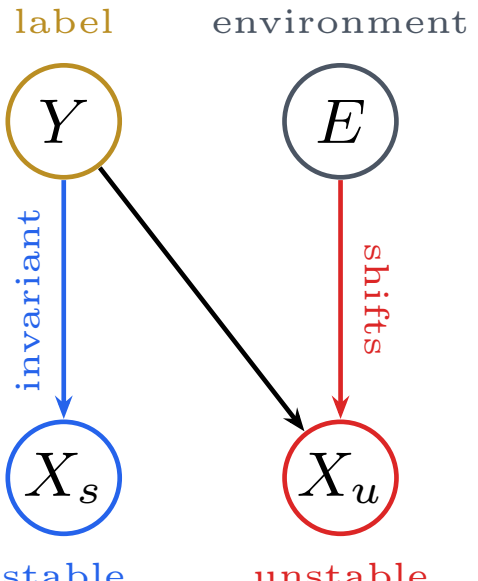```python
feature_ex = load_feature_extractor(trainable=False) # f_phi
train_net = initialize_model(trainable=True) # h_theta
for (images, labels) in dataloader: # (x_u, Y)
    optimizer.zero_grad()
    with torch.no_grad():
        features = feature_ex(images) # x_s
        inputs = torch.cat([images, features], dim=1) # z
        inputs = F.dropout3d(inputs, p=drop_prob) # \tilde{z}
    outputs = train_net(inputs)
    loss = loss_fn(outputs, labels)
    loss.backward()
    optimizer.step()
```